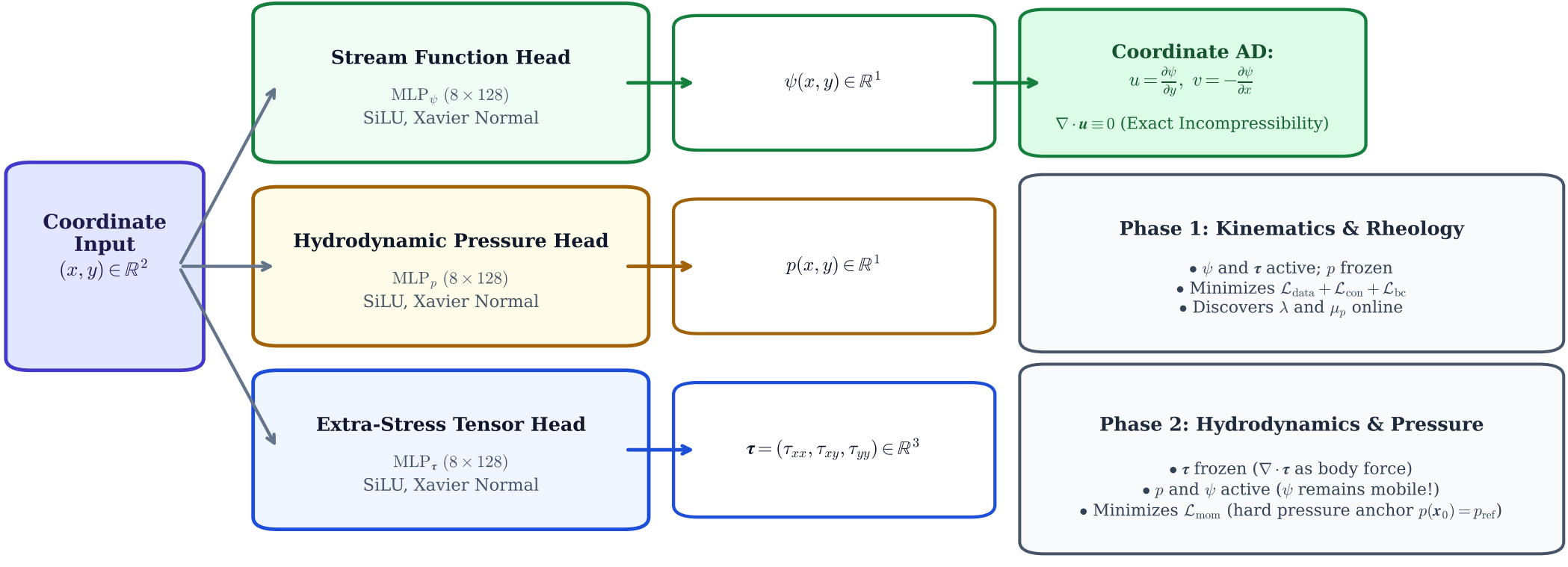

Coordinate Input
$(x, y) \in \mathbb{R}^2$
Stream Function Head
MLP_ψ (8×128)
SiLU, Xavier Normal
$\psi(x, y) \in \mathbb{R}^1$
Coordinate AD:
$u = \frac{\partial \psi}{\partial y},\ v = -\frac{\partial \psi}{\partial x}$
$\nabla \cdot \boldsymbol{u} \equiv 0$ (Exact Incompressibility)
Hydrodynamic Pressure Head
MLP_p (8×128)
SiLU, Xavier Normal
$p(x, y) \in \mathbb{R}^1$
Extra-Stress Tensor Head
$\mathrm{MLP}_{\boldsymbol{\tau}}$ (8×128)
SiLU, Xavier Normal
$\boldsymbol{\tau} = (\tau_{xx}, \tau_{xy}, \tau_{yy}) \in \mathbb{R}^3$
Phase 1: Kinematics & Rheology
• ψ and $\boldsymbol{\tau}$ active; p frozen
• Minimizes $\mathcal{L}_{\mathrm{data}} + \mathcal{L}_{\mathrm{con}} + \mathcal{L}_{\mathrm{bc}}$
• Discovers λ and μ_p online
Phase 2: Hydrodynamics & Pressure
• $\boldsymbol{\tau}$ frozen ($\nabla \cdot \boldsymbol{\tau}$ as body force)
• p and ψ active (ψ remains mobile!)
• Minimizes $\mathcal{L}_{\mathrm{mom}}$ (hard pressure anchor $p(\boldsymbol{x}_0) = p_{\mathrm{ref}}$)